

Retinal and anterior segment pathology

Pre-clinicals | Connect the tissue, lesion and visual consequence

Read the layer and circulation

Two retinal blood supplies

The central retinal artery supplies the inner retina, including ganglion cells and their axons. The choriocapillaris, supplied by posterior ciliary arteries, supports the outer retina and photoreceptors. The retina needs both circulations; preserved choroidal perfusion cannot fully replace central retinal artery perfusion.

Artery versus vein

Central retinal artery occlusion	Central retinal vein occlusion
Inner retinal ischaemia and swelling	Impaired venous drainage, congestion and leakage
Pale retina with a cherry red fovea	Widespread haemorrhages and dilated, tortuous veins
Foveal region stays relatively red because of underlying choroidal perfusion	Macular oedema can impair central acuity

Fundus lesion → mechanism

Finding	Meaning
Cotton wool spot	Nerve fibre layer ischaemia with interrupted axoplasmic transport
Hard exudate	Lipid/protein deposits from chronic vascular leakage
Flame shaped haemorrhage	Blood tracks along superficial nerve fibre bundles
Dot/blot haemorrhage	Blood within deeper retinal layers

Localise the visual consequence

- Macular injury causes central blur, distortion or a central scotoma; peripheral vision may remain.
- A relative afferent pupillary defect suggests asymmetric afferent injury, such as optic nerve damage or extensive retinal ischaemia.

Diabetes leakage and ischaemia

One disease, two major pathways

- Chronic hyperglycaemia injures retinal capillaries and pericytes
↓
- Barrier breakdown causes leakage; capillary closure causes ischaemia
↓
- Leakage produces oedema and hard exudates
↓
- Ischaemia increases VEGF and can drive new vessel formation

Know what makes it proliferative

Feature	Interpretation
Microaneurysms, dot/blot haemorrhages, hard exudates	Non-proliferative microvascular disease
Cotton wool spots, venous changes, intraretinal microvascular abnormalities	Evidence of retinal ischaemia; new surface vessels need not be present
New vessels at disc or elsewhere on retina	Defines proliferative diabetic retinopathy
Retinal thickening at the macula	Diabetic macular oedema; may occur at different retinopathy stages

Why new vessels are dangerous

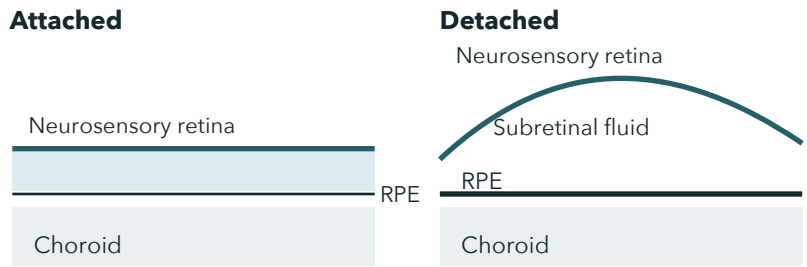
- Fragile neovessels can bleed into the preretinal space or vitreous, obscuring the optical path.
- Associated fibrovascular membranes can contract and pull the neural retina off the pigment epithelium: tractional detachment.
- VEGF driven iris and angle neovascularisation can impair aqueous drainage and cause neovascular glaucoma.
- Central vision is also vulnerable to macular capillary non-perfusion. Not all diabetic central blur is caused by leakage.

Three examination traps

Cotton wool spots are not lipid deposits. Macular oedema is not restricted to proliferative disease. Neovascularisation and intraretinal microvascular abnormalities are different findings, so do not classify every abnormal vascular pattern as proliferative retinopathy.

Separation, traction and macular injury

Retinal detachment plane



Retina separates from RPE; fluid fills the subretinal space.

Three routes to the same separation

Type	Mechanism
Rhegmatogenous	Full thickness retinal break permits liquefied vitreous to enter the subretinal space
Tractional	Contracting fibrovascular tissue pulls on the retina; a break is not required
Exudative / serous	Fluid accumulates because of leakage or impaired fluid handling; no primary break or traction is required

From ageing vitreous to a retinal break

- Vitreous liquefies and posterior vitreous separates
- ↓
- Persistent focal adhesions create traction
- ↓
- Traction can cause flashes, a tear or haemorrhage
- ↓
- A tear can permit fluid entry and rhegmatogenous detachment

High yield distinctions

- Posterior vitreous detachment separates vitreous from retina. Retinal detachment separates neural retina from RPE.
- New floaters and flashes with a curtain-like field loss suggest retinal detachment; it is usually painless.
- Age-related macular degeneration affects central vision. Drusen lie between the RPE and Bruch membrane.
- Neovascular AMD produces macular leakage or bleeding from abnormal choroidal vessels, causing central distortion and subretinal fluid.

Cornea, lens and anterior uvea

Localise the loss of clarity

Corneal endothelial pump failure allows stromal swelling and light scattering. Lens protein and fibre changes cause cataract. Anterior uveal inflammation releases cells and protein into aqueous; ciliary spasm contributes to pain.

Cataract patterns worth retaining

Pattern	High yield clue
Nuclear sclerosis	Increasing nuclear refractive index can cause a myopic shift and temporary improvement in unaided near vision
Cortical	Hydration changes in lens fibres produce radial spoke-like opacities and glare
Posterior subcapsular	Central posterior opacity often causes glare; associated with corticosteroid exposure

Anterior uveitis to secondary pressure elevation

- Blood-aqueous barrier breaks down: inflammatory cells plus protein flare
- Inflamed iris adheres to lens: posterior synechiae
- Circumferential adhesions can obstruct pupillary aqueous flow
- Posterior chamber pressure rises, iris bows forward and the angle can close

Mechanism checkpoints

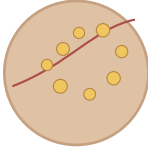
- Cells are leukocytes; flare is light scattering by protein in the aqueous. They are separate signs.
- Keratic precipitates are inflammatory deposits on the corneal endothelium.
- Presbyopia is loss of accommodation from reduced lens flexibility. Improved near vision during nuclear cataract does not restore accommodation.
- Anterior uveitis may cause a painful, photophobic red eye with a small or irregular pupil. Acute angle closure more often produces a mid-dilated poorly reactive pupil with a markedly raised IOP.

Visual pathology and references

Original educational schematics. Compare the lesion and its layer, then explain the mechanism.

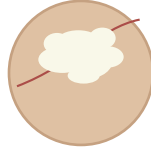
Four fundus clues

HARD EXUDATES



Sharp yellow deposits within the retina

COTTON WOOL SPOT



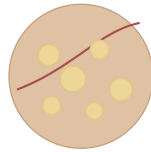
Fluffy pale patch nerve fibre layer

HAEMORRHAGES



Flame: superficial
Dot/blot: deeper retina
Hard exudates reflect leakage; cotton wool spots reflect superficial ischaemic axonal injury. Drusen occupy a different compartment below the RPE. [R1, R3]

DRUSEN

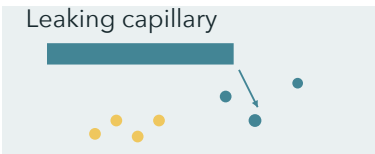


Deposits below RPE above Bruch membrane

Two pathways in diabetic retinopathy

CAPILLARY INJURY

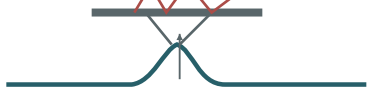
LEAKAGE



Macular oedema
and hard exudates

NON-PERFUSION

VEGF + new vessels



Fibrovascular traction
Fragile vessels may bleed

Leakage produces oedema. Non-perfusion drives VEGF, fragile neovessels and associated fibrovascular tissue; contraction may detach the retina. [R1, R2]

References for these figures

- R1. ADA, Management of Diabetes-Related Retinopathy (2019): retinal lesions and proliferative mechanisms.
- R2. National Eye Institute, Diabetic retinopathy: leakage, bleeding and traction.
- R3. Johnson et al., Drusen-associated degeneration in the retina (2003): the sub-RPE location.